Adil Sukumar

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EDUCATION

Vellore Institute of Technology **Aug'25–May'30**
Integrated M.Tech. Artificial Intelligence & Bioinformatics | CGPA: 9.10/10

EXPERIENCE

British Airways **Oct'26–Present**
Data Science Job Simulation

- Undertake a self-paced virtual data science simulation focused on **customer insights and predictive analytics** for airline business decisions.

Pawsitive Diagnostics **Feb'26–Present**
Founder and CEO

- Lead **software, hardware and machine learning** development for a **wearable animal-health system** integrating **motion, acoustic, environmental and surface-temperature sensing**.
- Build **edge-to-cloud workflows** for filtering, baselines, **activity recognition** and anomaly alerts.
- Develop **signal filtering**, **activity recognition** and context-aware **anomaly alerts** using individual animal baselines.

ElectraWireless **May'26–Aug'26**
Lead Intern, IoT & Embedded Systems

- Co-created **ELLY**'s **offline-first React/TypeScript** ecosystem with **Web-LLM** voice control, **TensorFlow.js/MobileNet** vision and **BLE/Wi-Fi** discovery.
- Automated **Capacitor** iOS and Android builds with **GitHub Actions**.
- Integrated Web BLE, Wi-Fi subnet discovery, **MQTT** and WebSockets for local device control and energy telemetry.

PROJECTS

Portiva | Java, Python, C++, Android, Arduino **2026**

- Built mission coordination, heading-aware **A* planning** and swept-body **collision checks** for a person-following delivery robot prototype.
- Integrated **Uno protocol**, **watchdog** and sensor-firmware workflows with an Android bench demo; physical motor outputs remain disabled pending commissioning.

SoftSpace and SoftFlex | CAD, Python, FEA **2026**

- Developed **six adjustable headphone CAD variants** and a glasses-aware cushion concept, retaining design history and traceable analysis evidence.
- Audited **eight higher-order joint reference cases** and mesh-displacement comparisons; documented material assumptions and required physical tests.

LightSense Connected Lighting | ESP32, WS2812B, MQTT **2026**

- Developed **ESP32** firmware and backend workflows for separate RGB and health-status LEDs, with retained **MQTT** commands for state restoration after reconnect.

TECHNICAL SKILLS

Embedded and Languages: Arduino, ESP32/ESP32-C3, C/C++, Python, Java, Android
Connectivity: BLE, Wi-Fi, MQTT, WebSockets, serial protocols, REST APIs, FastAPI
IoT and Sensing: Sensor integration, telemetry, signal filtering, baseline modelling, anomaly alerts
Edge AI and Robotics: TensorFlow.js, MobileNet, Web-LLM, ONNX, A* planning, collision checks
Design and Tools: CAD, FEA, mesh comparison, Git, Linux, Docker, GitHub Actions, Capacitor

LEADERSHIP AND ACHIEVEMENTS

VIT Student Council — Member Secretary – Technical | Sep'26–Present
WiCyS Student Chapter, VIT — Secretary | Dec'25–Present

- CyberShield Hackathon 2025, VIT — **First Place**.
- SKITECH National Innothon 3.0, Indore — **First Runner-Up**, national level.
- SASI National Hackathon 2026, Andhra Pradesh — **Top 6**, national level.